DNA Structure and DNA Polymerase

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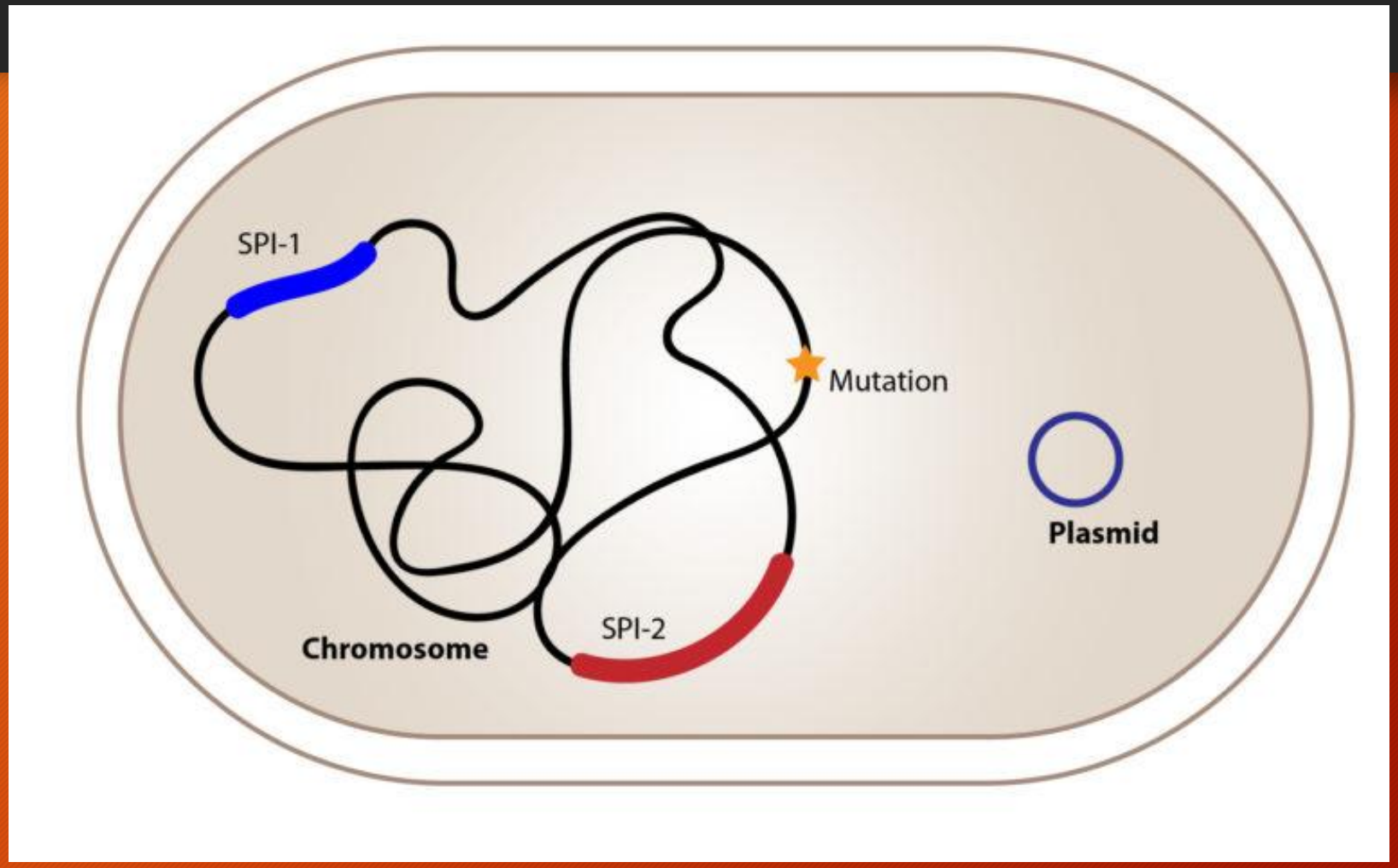
Jan 27 2026

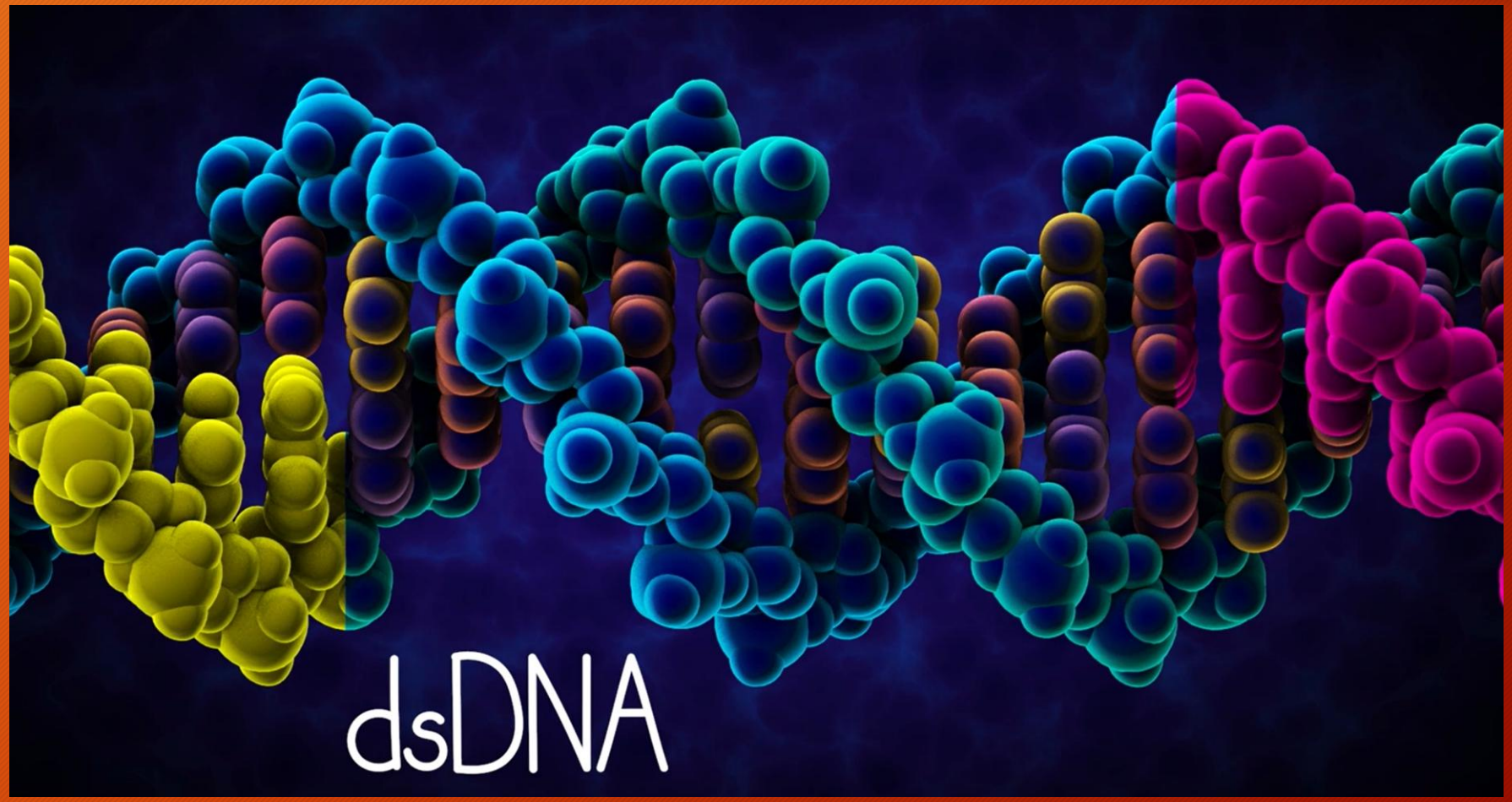
Prokaryotes from Genome to Proteins



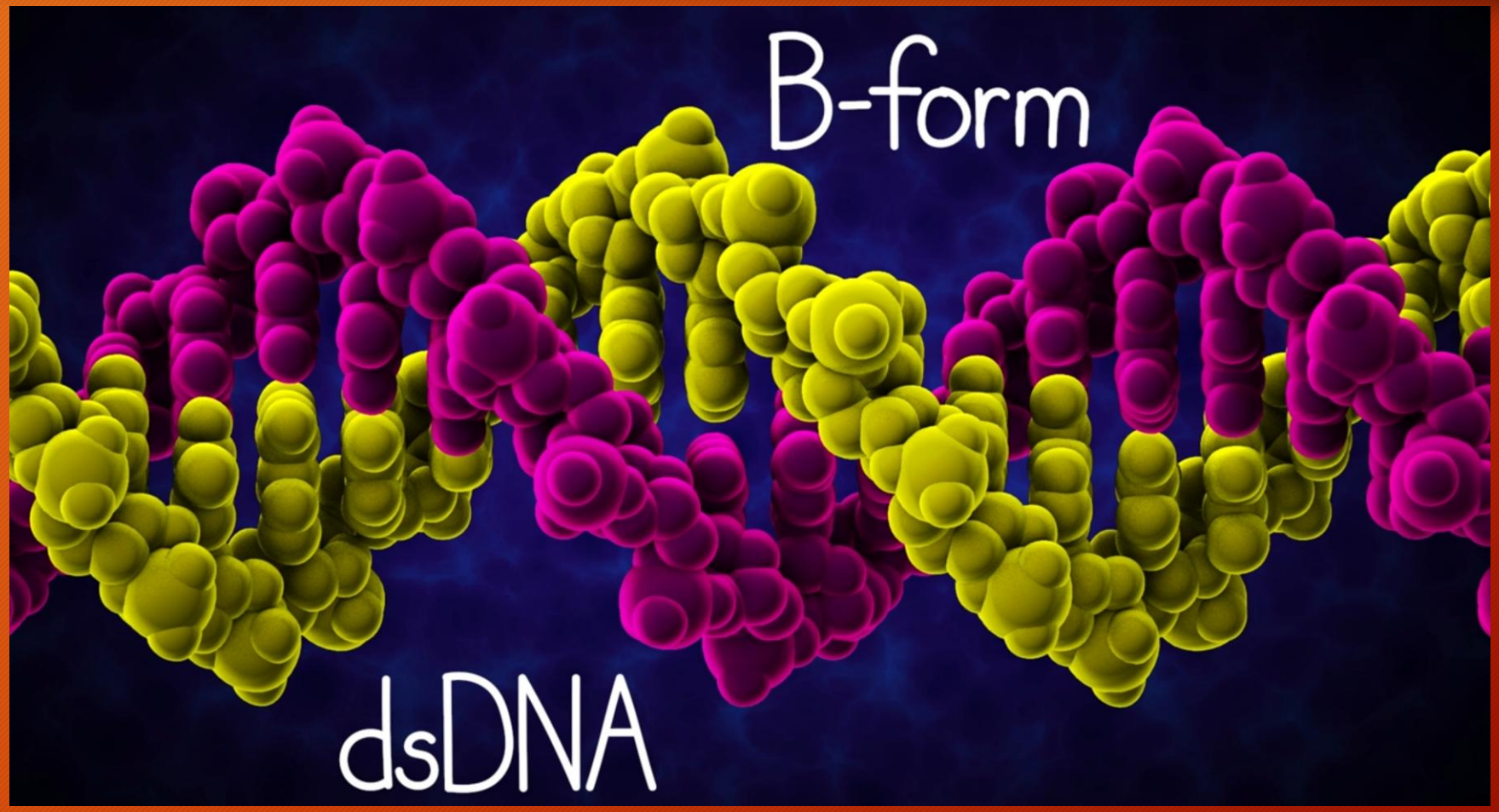
Goals and Objectives

- Goals:
  - Understand how DNA polymerase catalyzes the addition of deoxynucleoside triphosphates (dNTPs) to a growing DNA strand.
  - Learn experimental assays for detecting and characterizing DNA polymerase activity.
- Objectives: After completing the materials for this unit, you should be able to:
  - Recognize the structure of nucleic acids and key features of the structure of DNA polymerase.
  - Describe the factors that confer high accuracy to DNA polymerase activity.
  - Predict if a DNA-binding protein binds to the major or minor groove of DNA given the protein function.
  - Summarize how critical residues of DNA polymerase and metal ions facilitate catalysis.
  - Illustrate how a tautomeric state will result in the addition of the wrong base.
  - Select the best experiment to test DNA polymerase activity for a given hypothesis and experimental constraints.
  - Define processivity in both general terms and specifically in regards to DNA polymerase.
  - Troubleshoot negative results from a hypothetical DNA polymerase processivity assay.
  - Interpret data from DNA replication experiments.
  - Analyze protein structures to infer functional information.

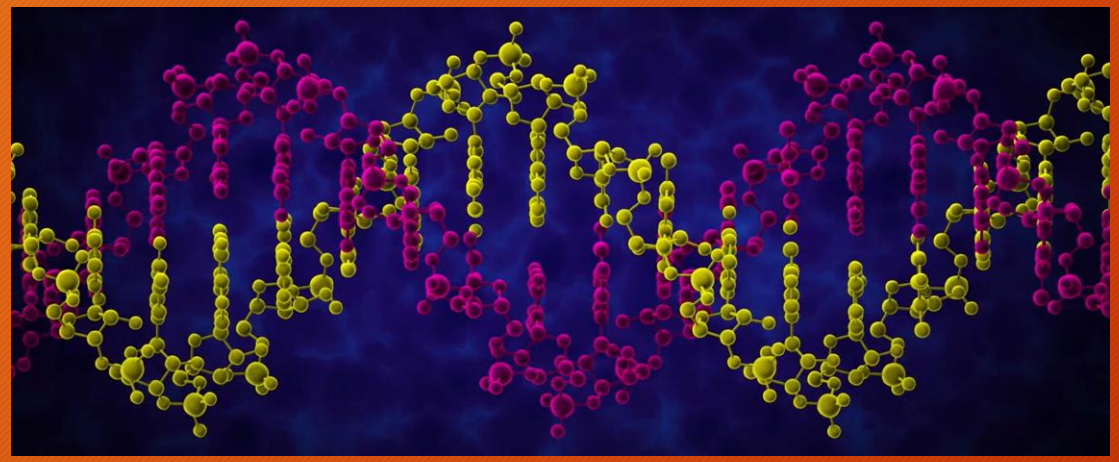
Double strand DNA (dsDNA)

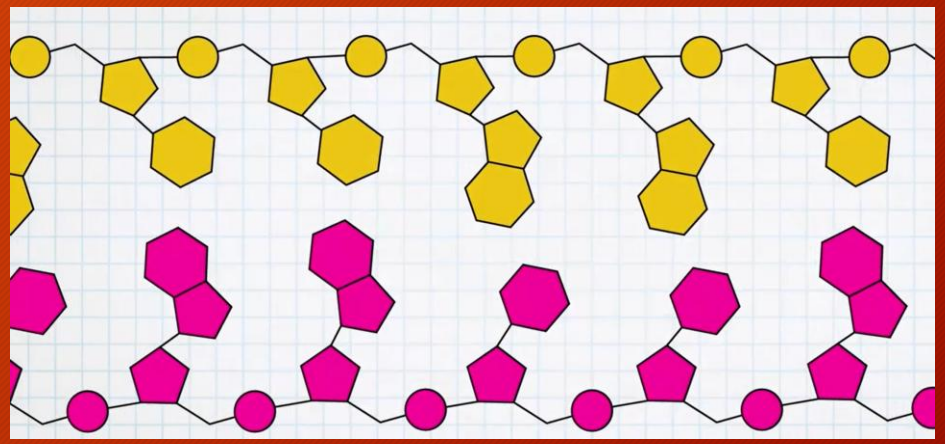


B-form DNA

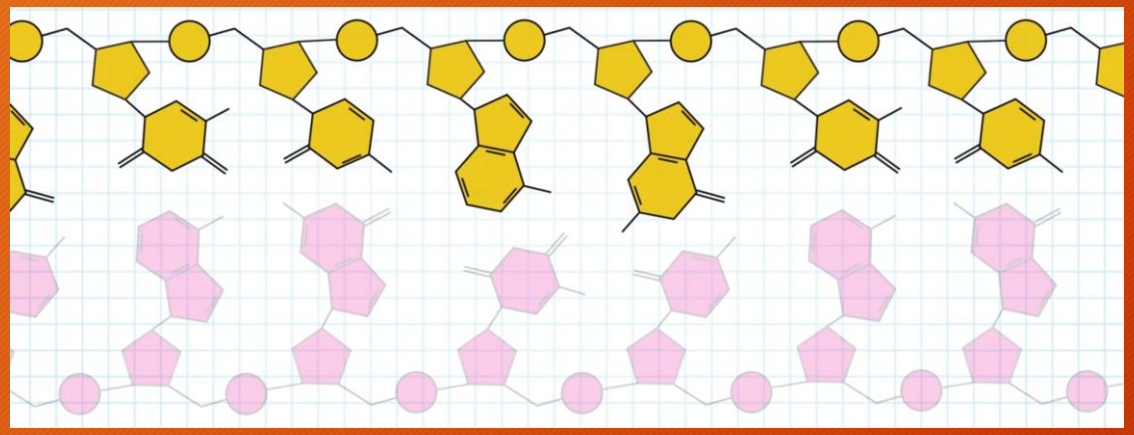


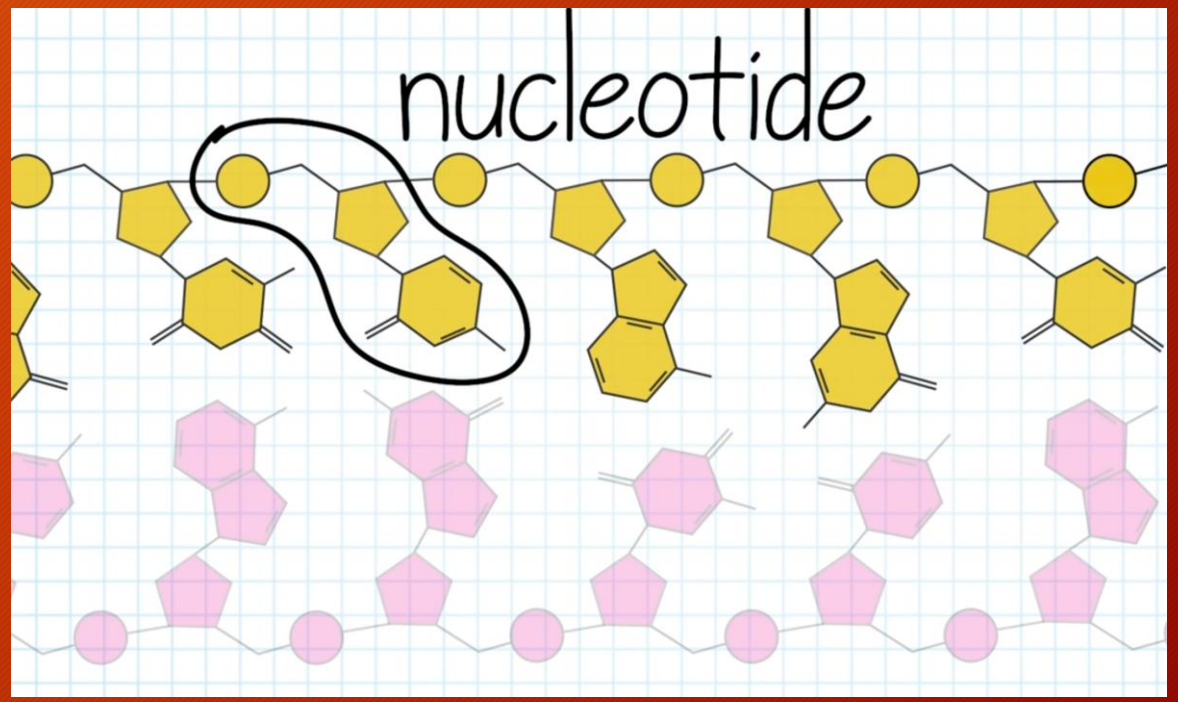
Simplification version of DNA



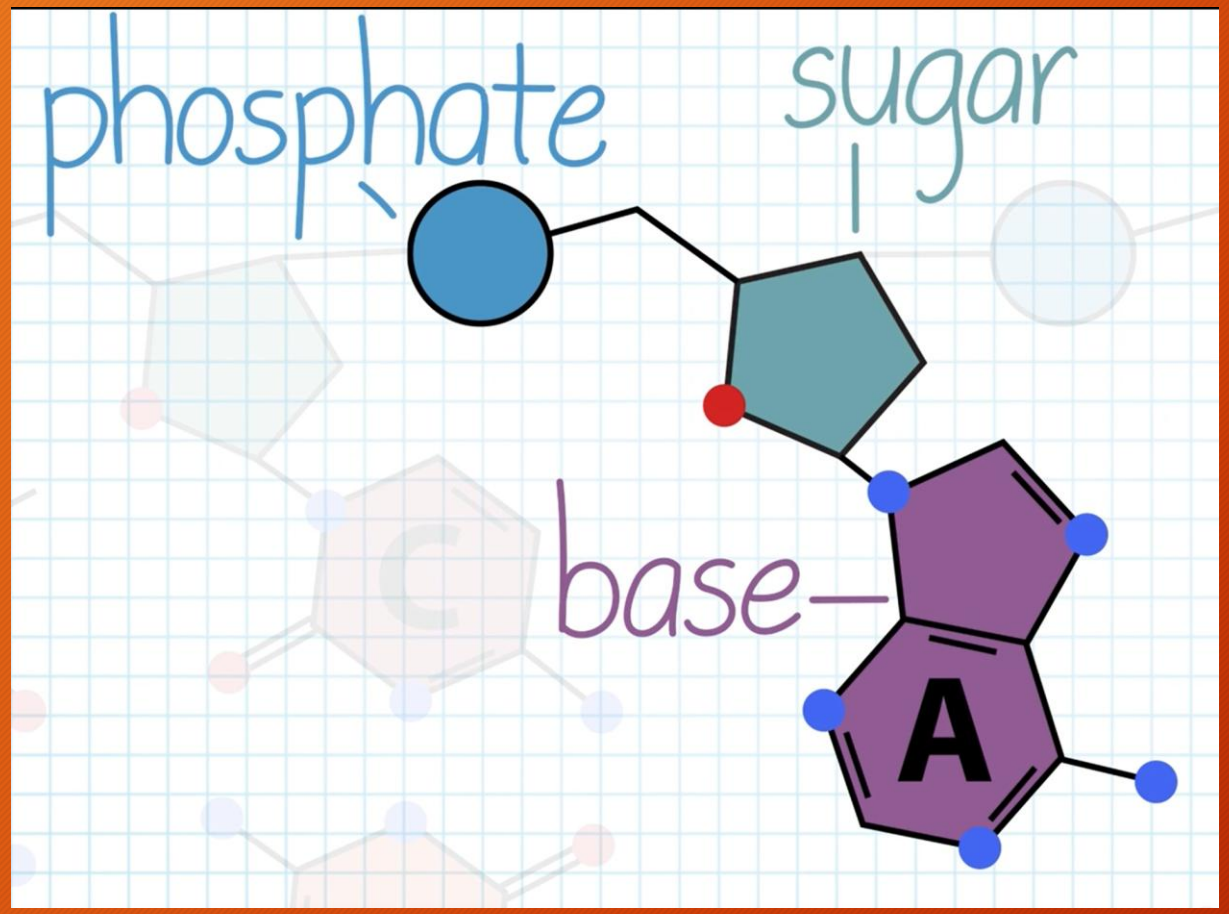


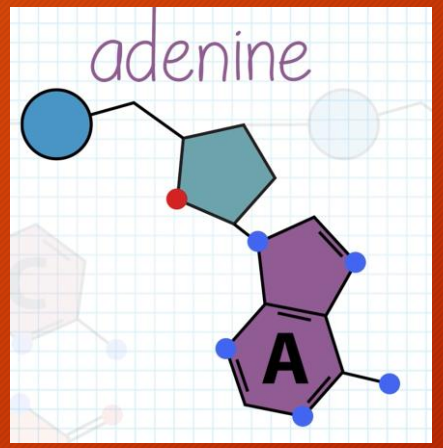
Each strand of DNA is a poly nuclotide

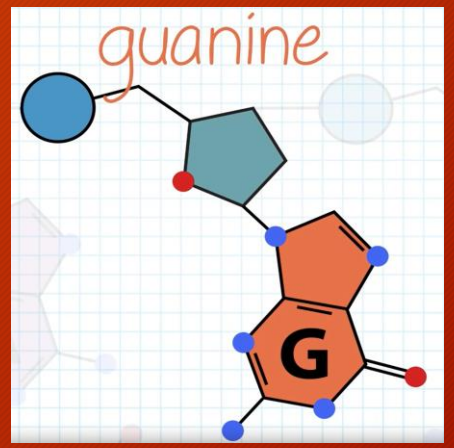


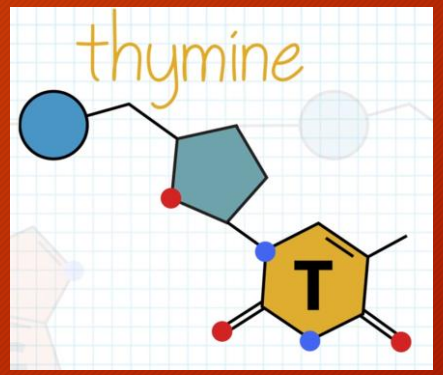


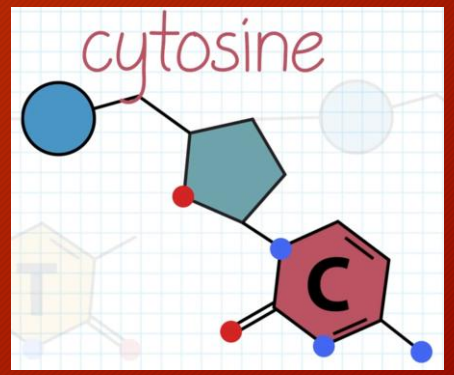
Nucleotide = Phosphate-sugar-base



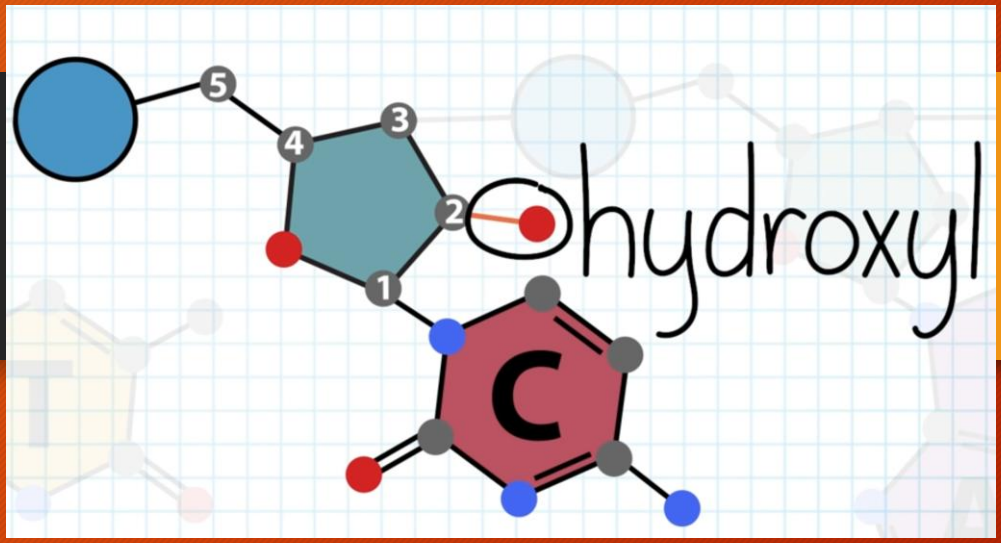


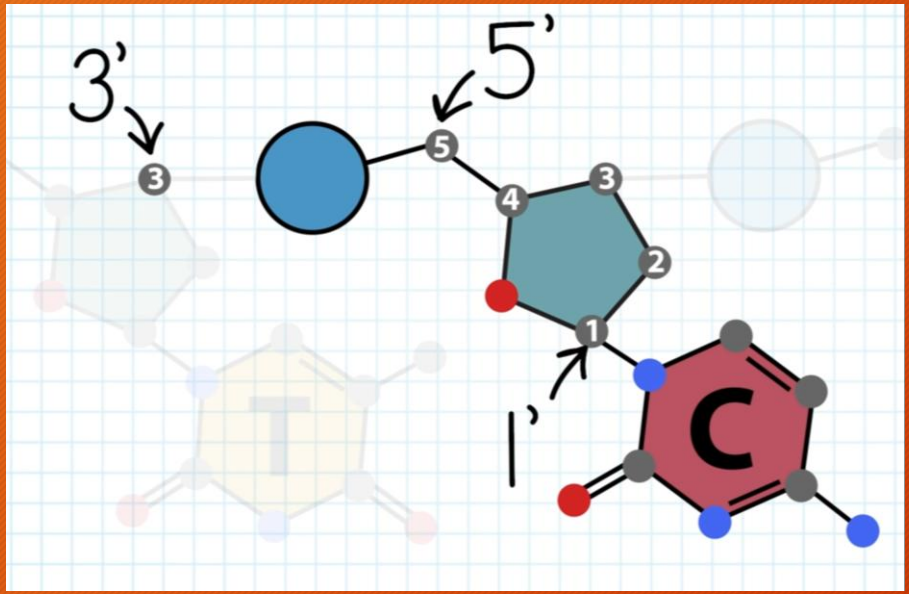


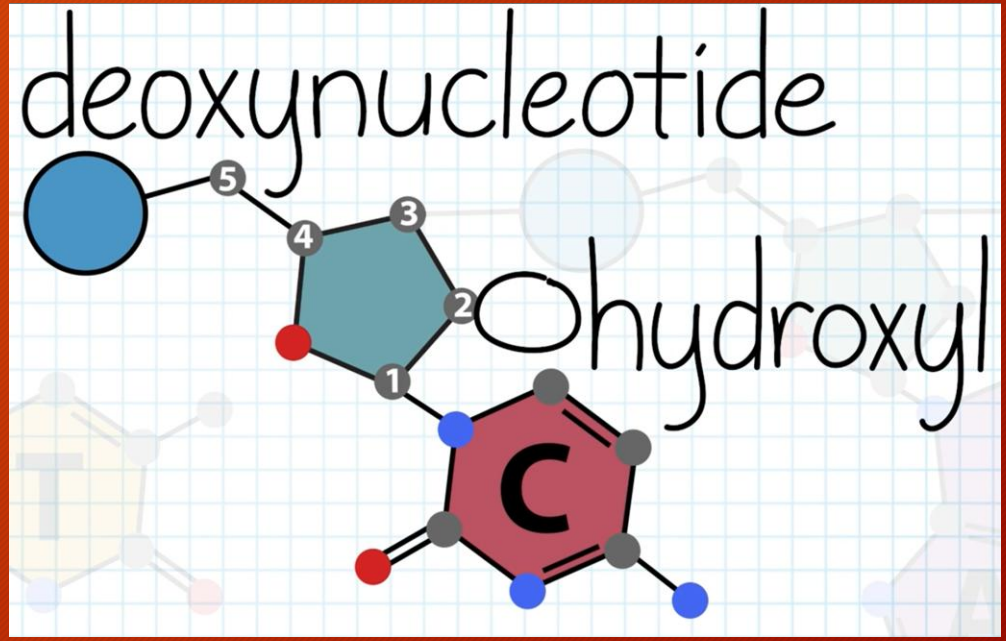




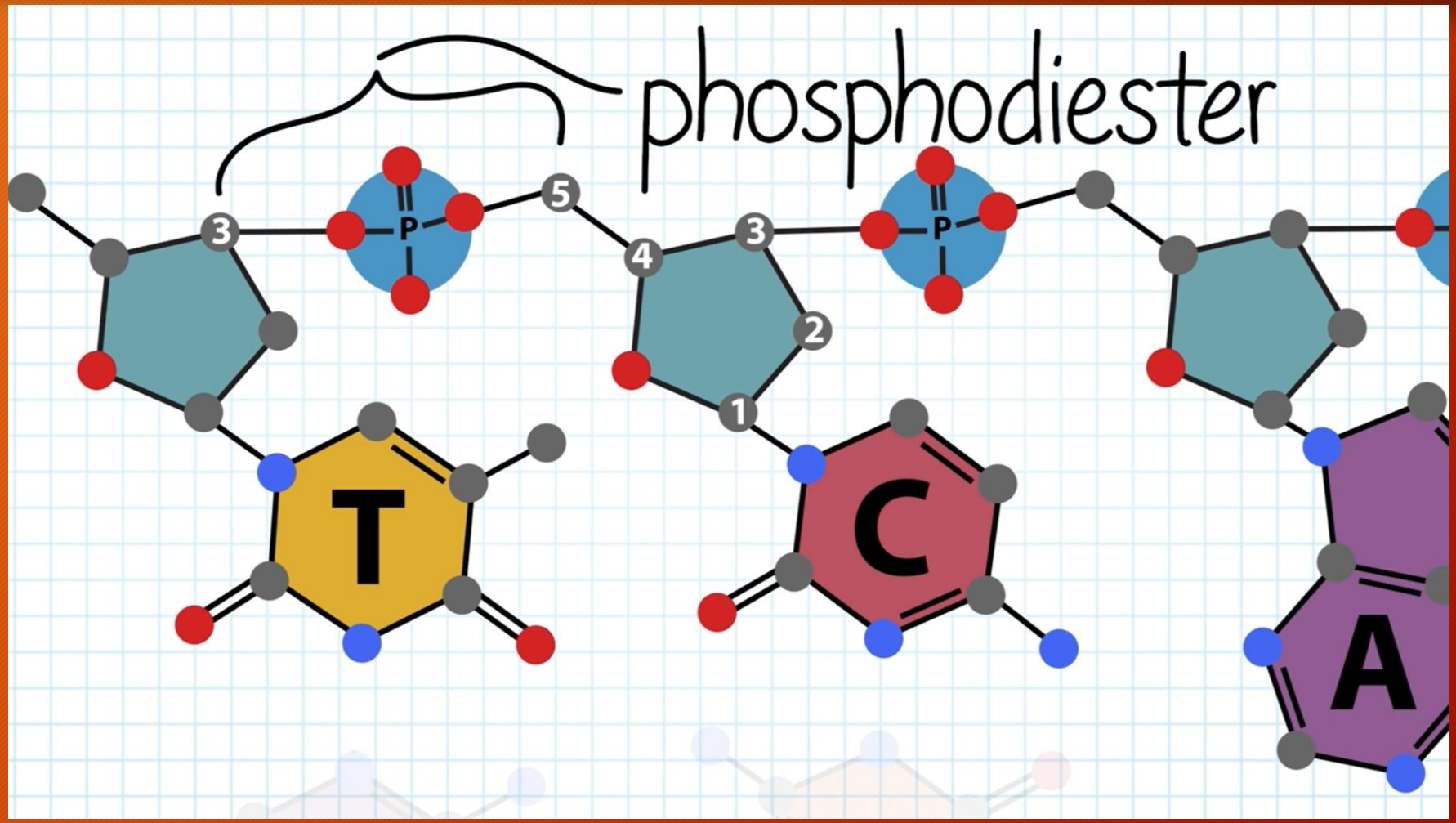
DeoxyNucleotide



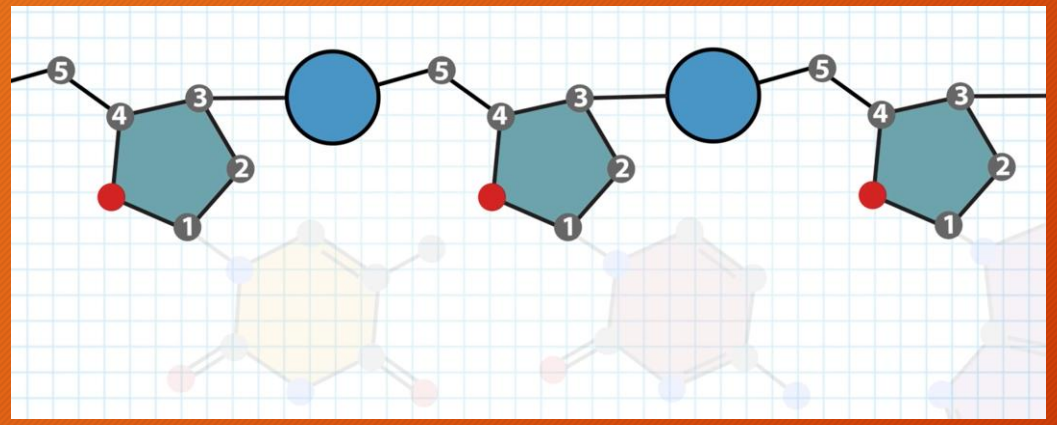


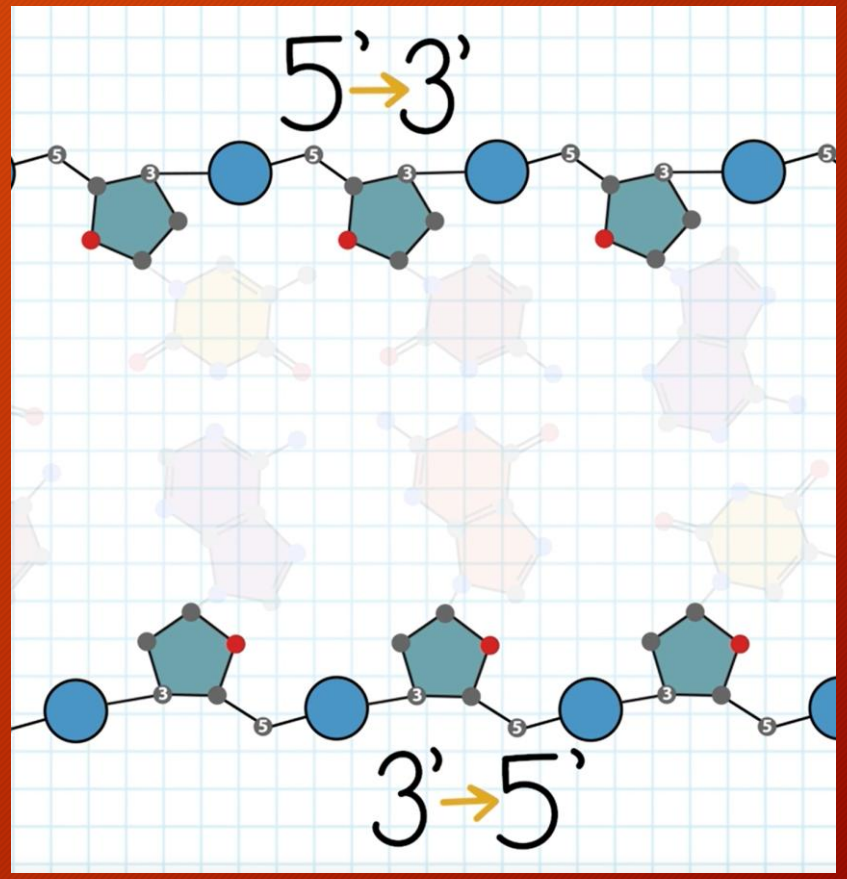


Phosphodiester bonds T and C



Phosphate and Sugar is the backbone of DNA





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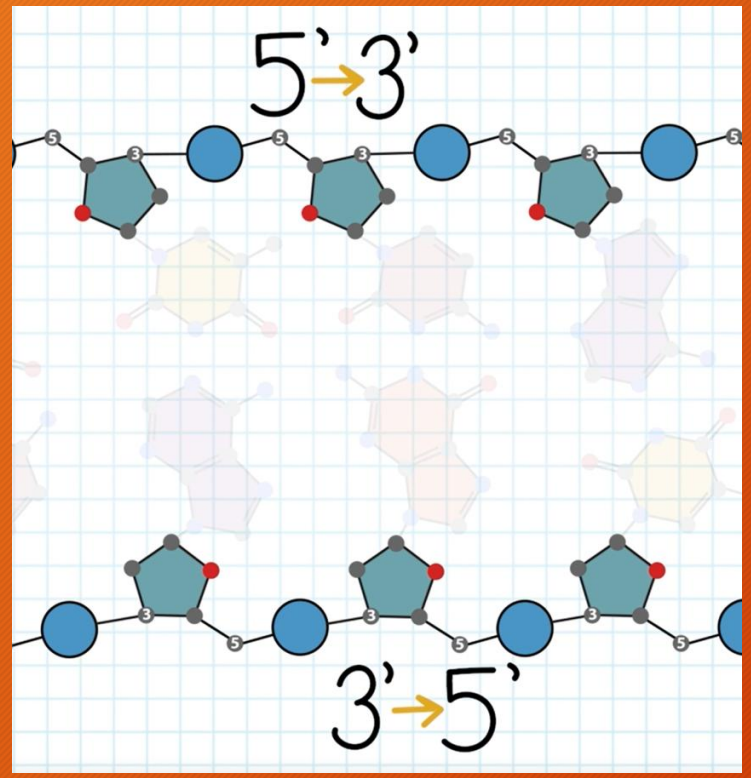

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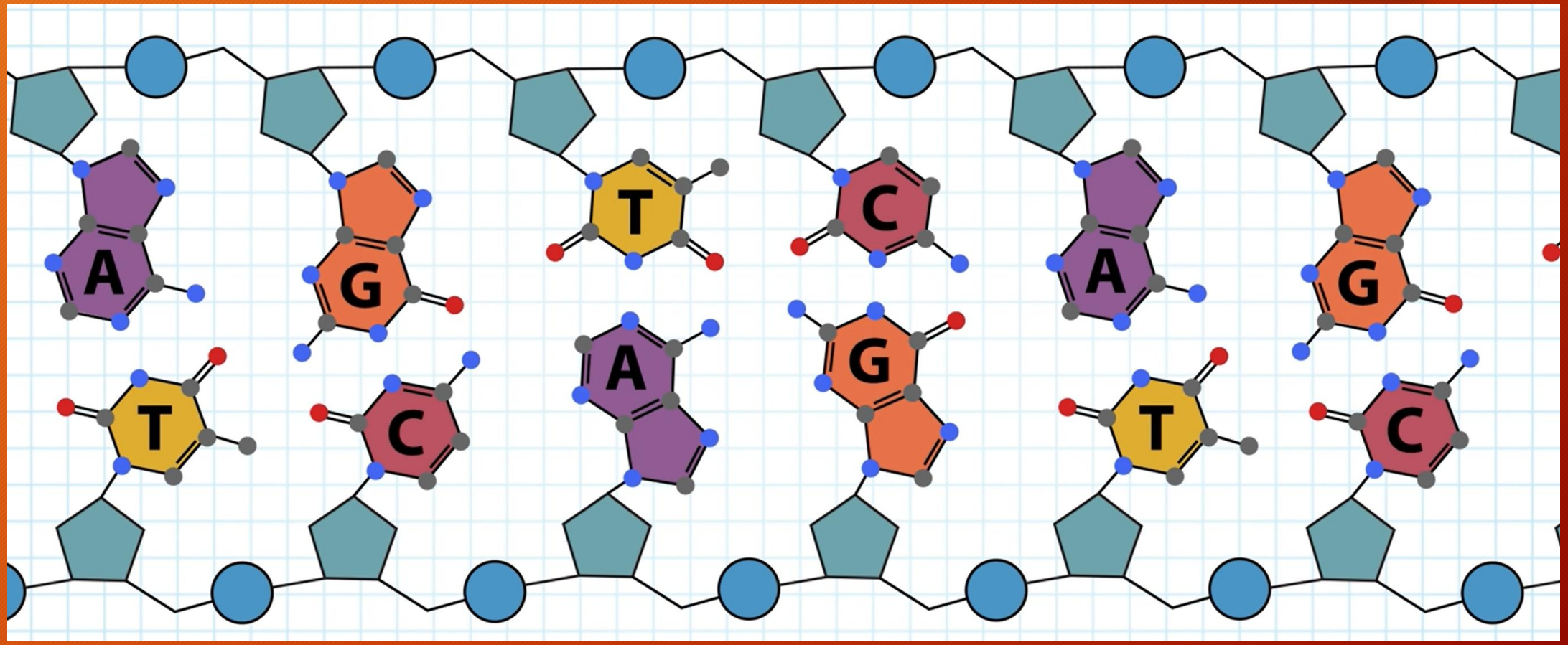
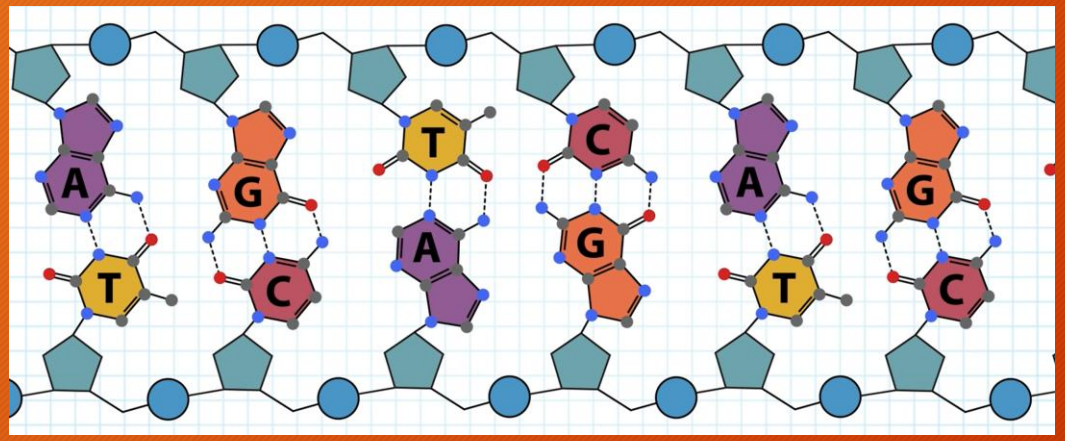
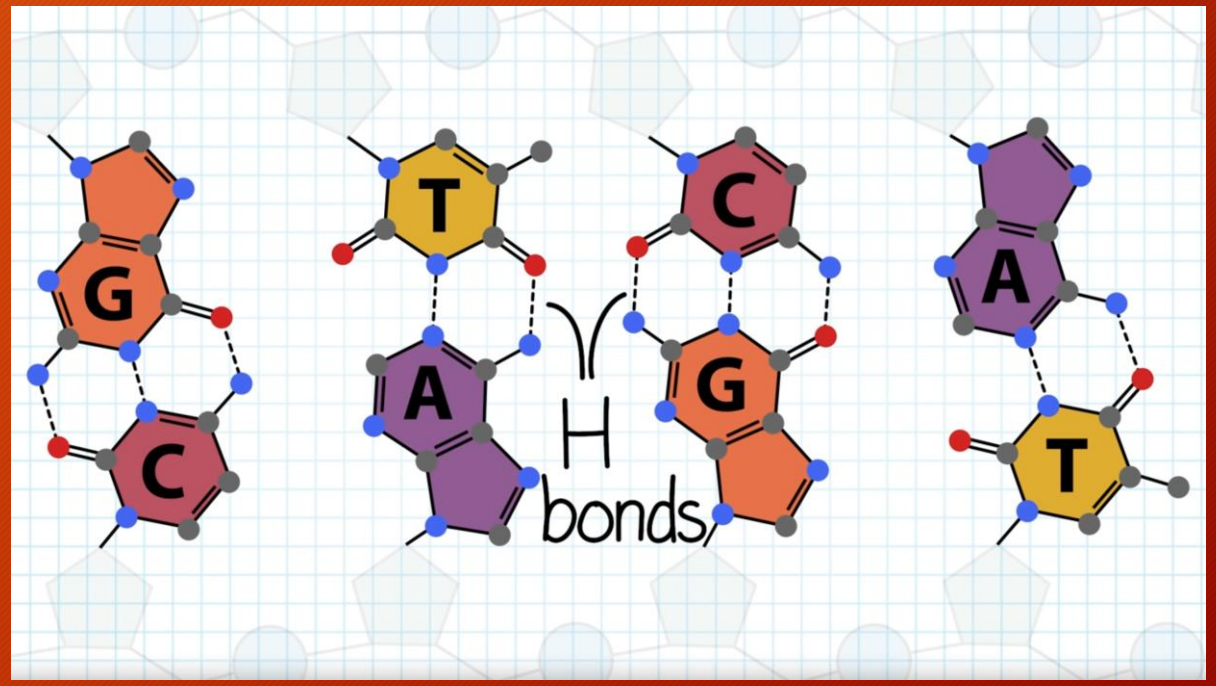
This is NOT the real structure of DNA

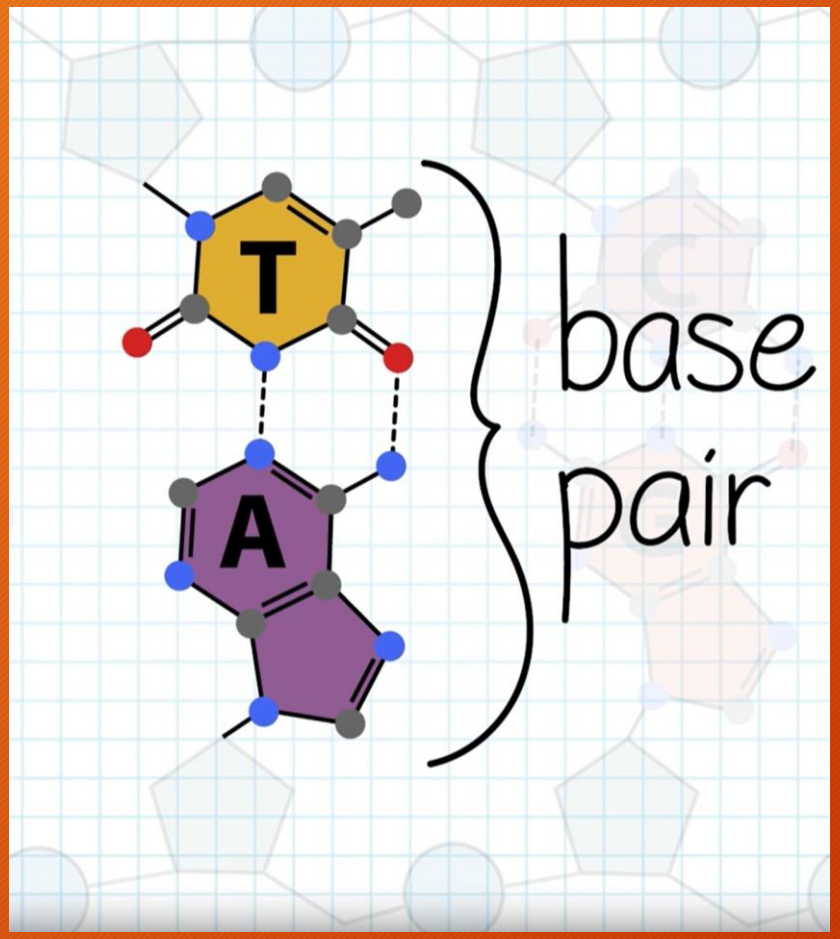


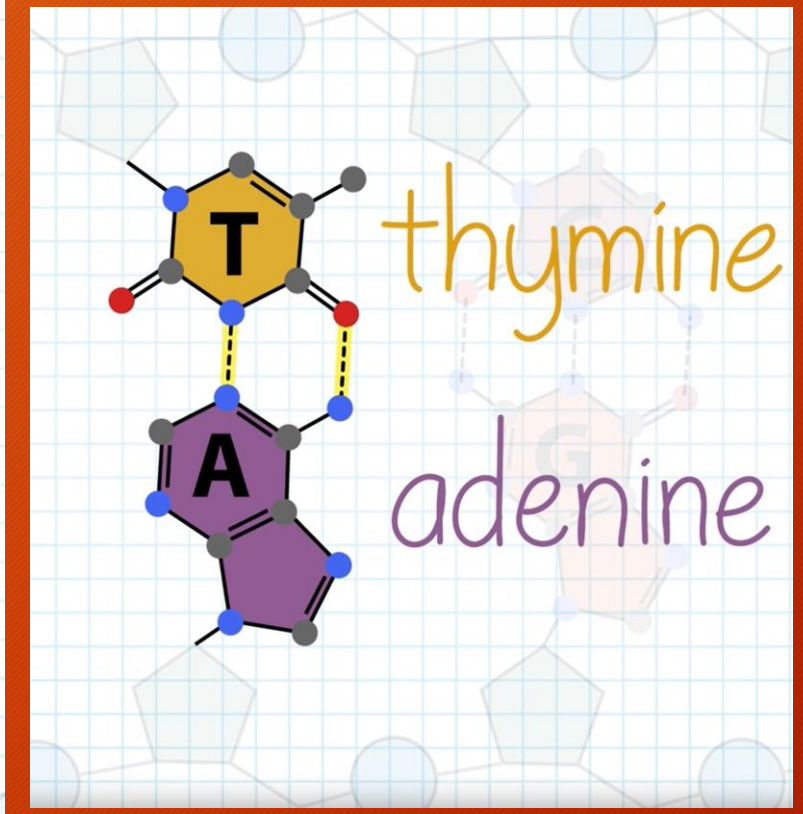
Hydrogen bonds

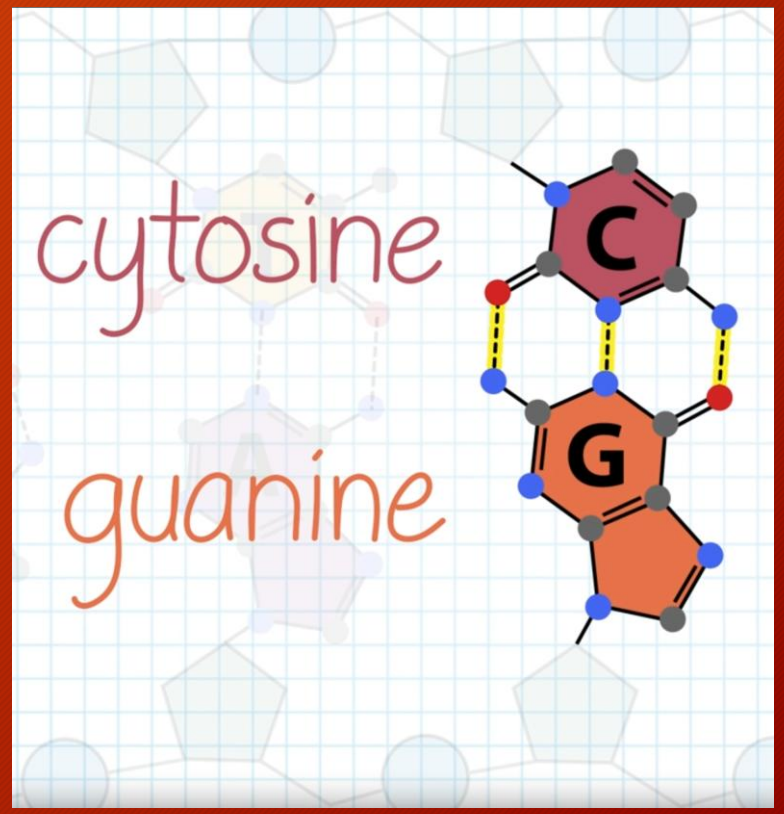




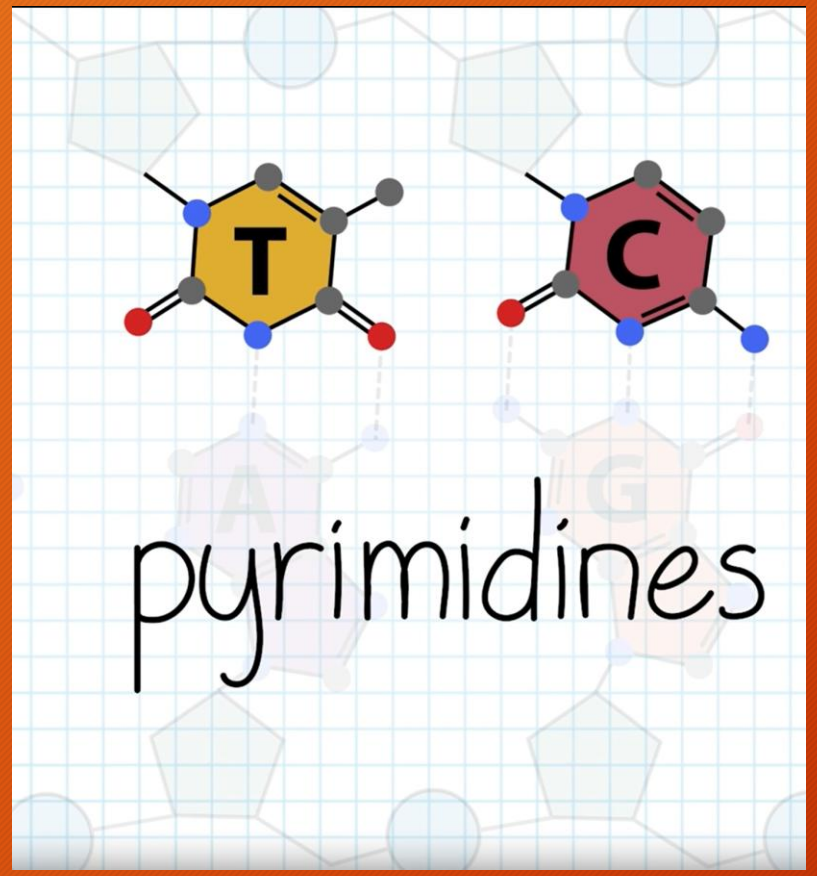
Base pair

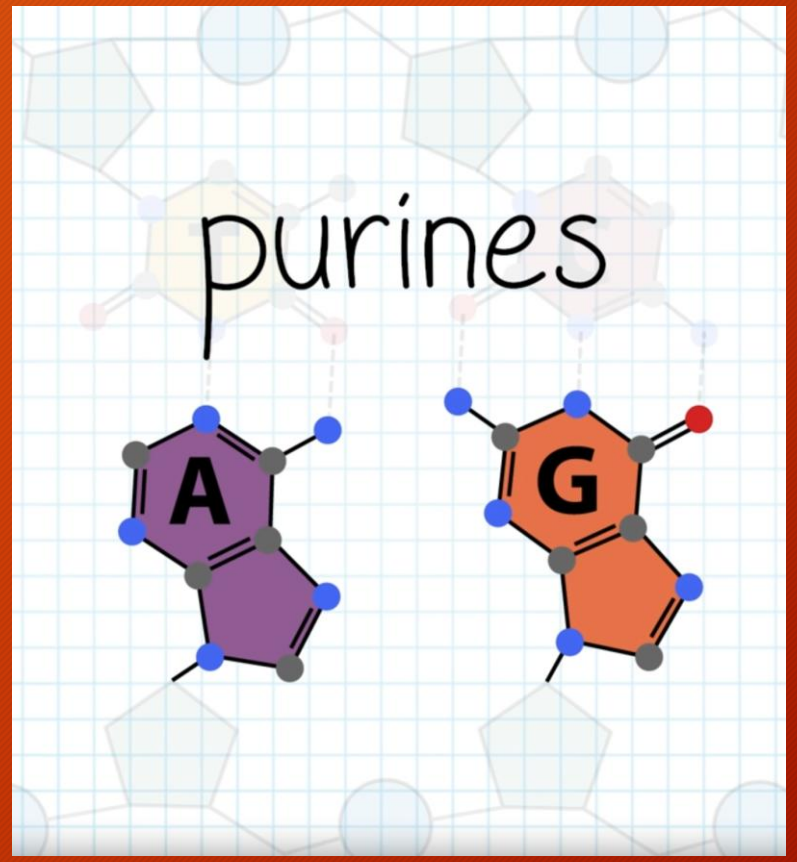




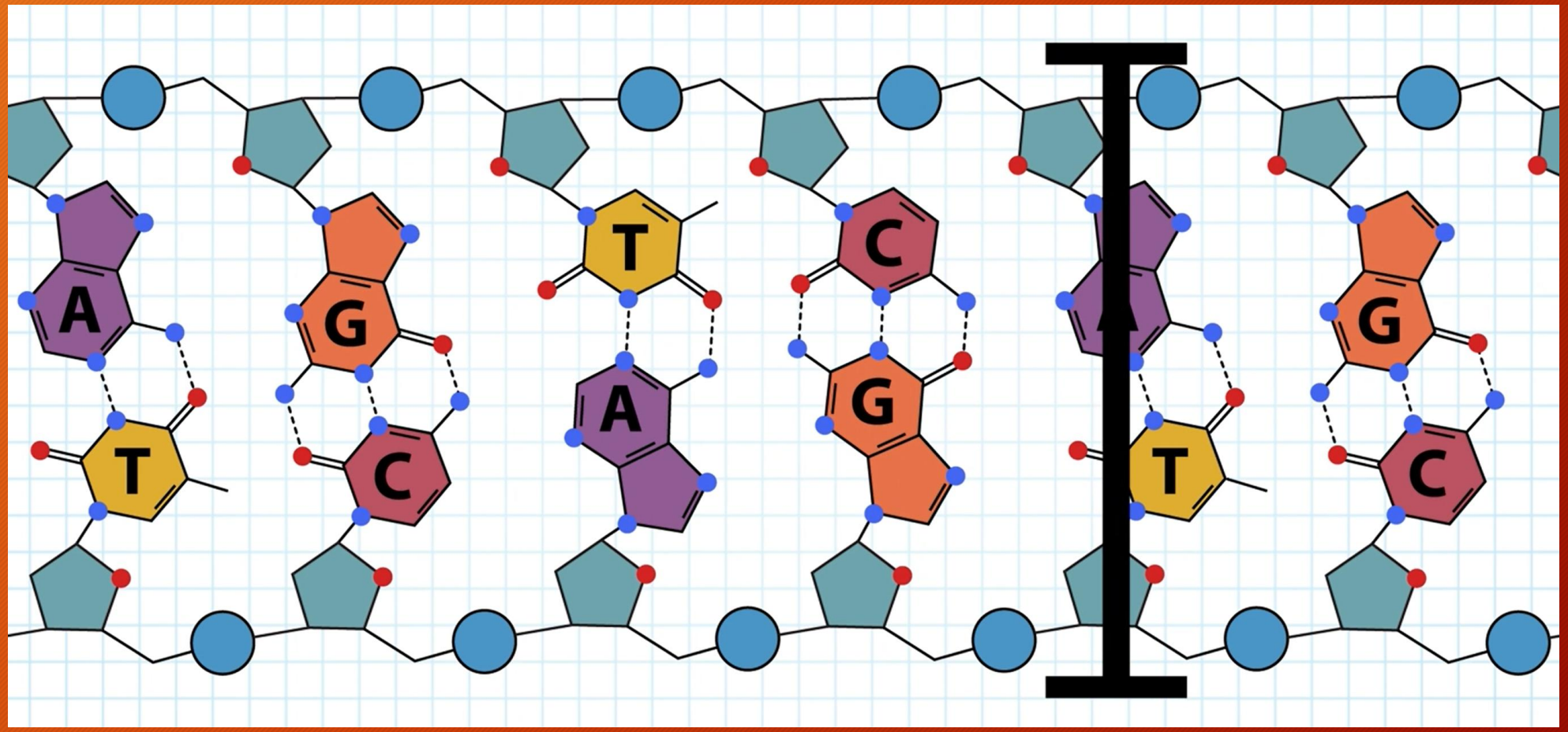


Pyrimidines vs Purines

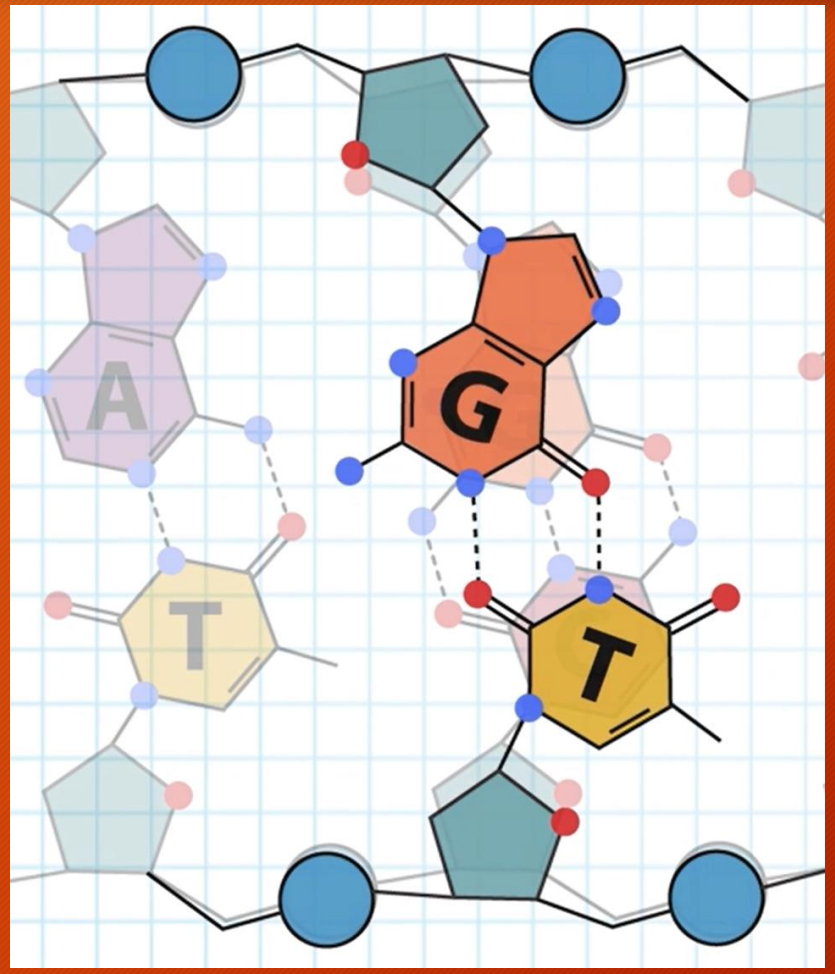




Distance is the same

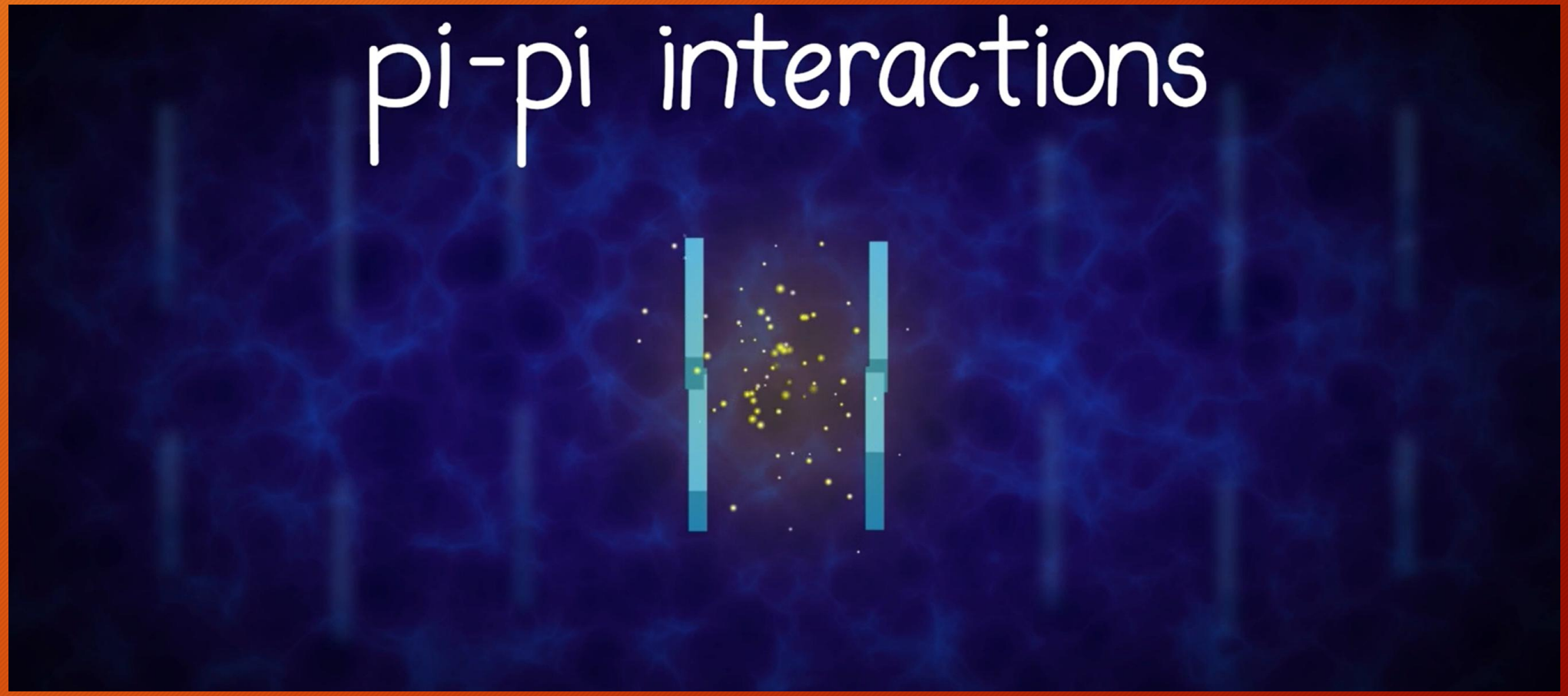


G and T are NOT the same geometry

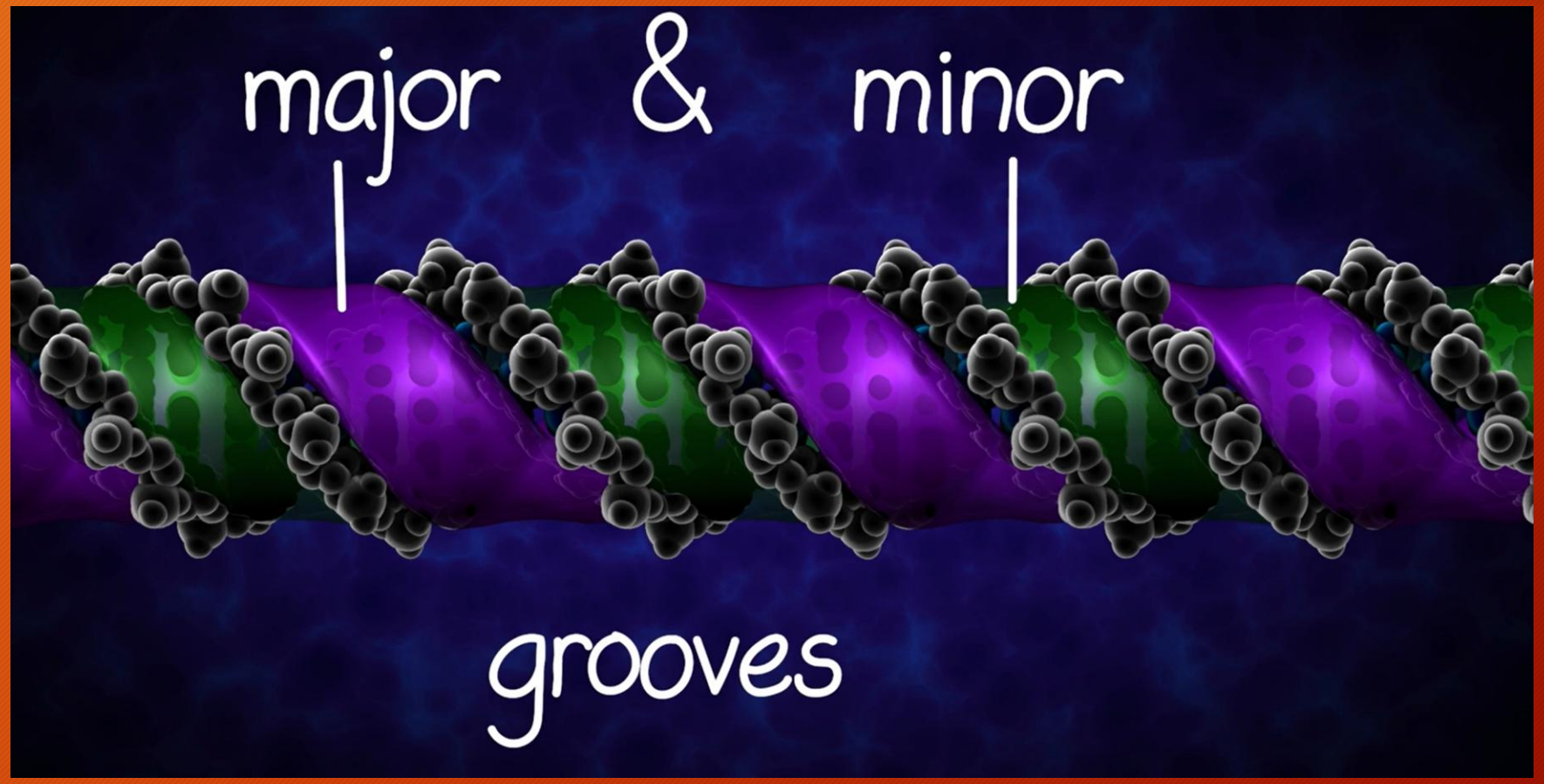


1 turn = 10 base pair

Pi-Pi Interactions



Major and minor grooves



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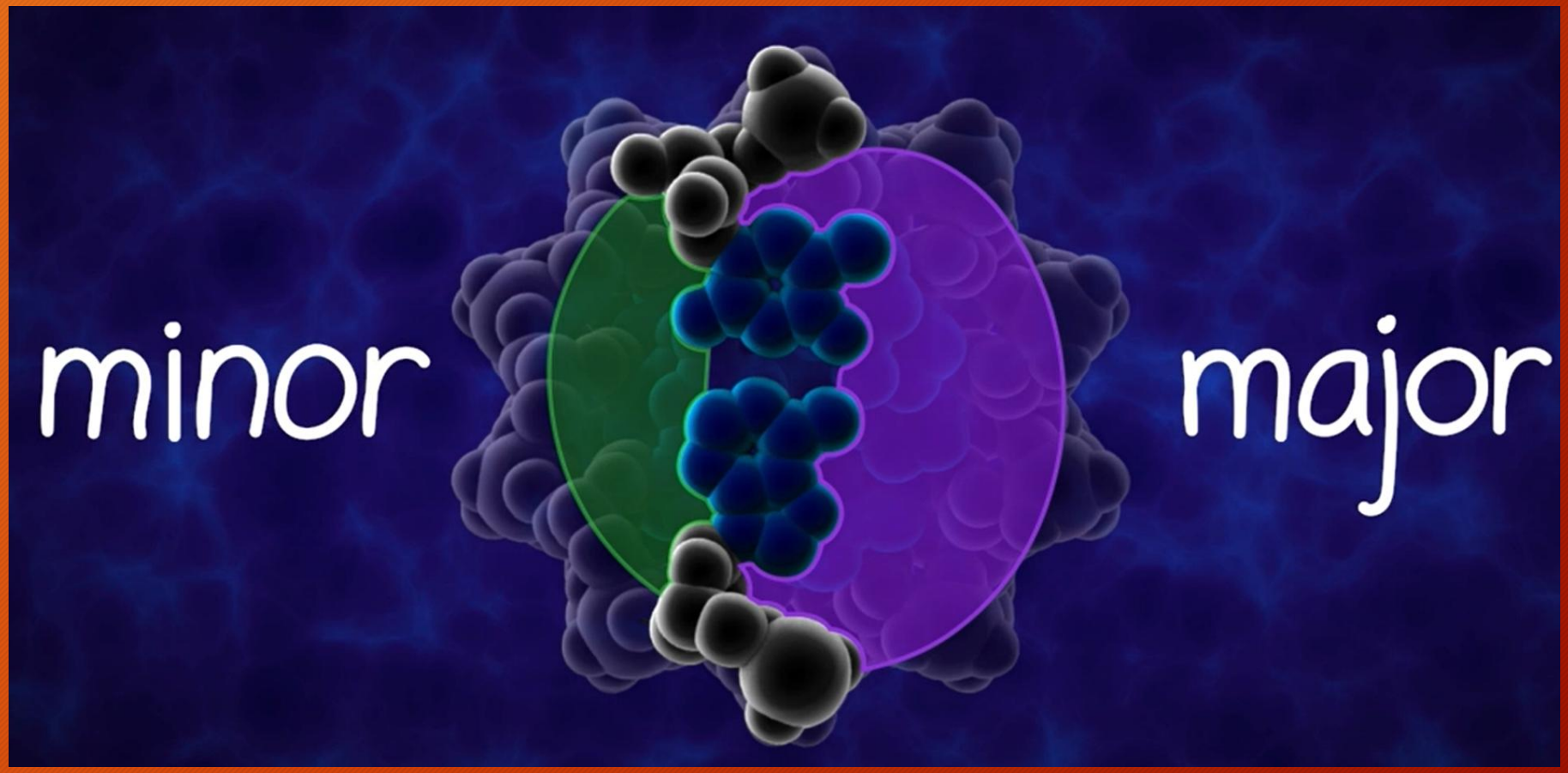

Select the options to complete the sentences describing the bonds in DNA structure.

The backbone of a DNA strand is comprised of (1)

a) Covalent

b) Non-covalent

c) Ionic

bonds between sugar and phosphate groups.

Two DNA strands interact through (2)

a) Covalent

b) Non-covalent

c) Ionic

hydrogen bonds between bases, in what is called (3)

a) base pairing.

b) base stacking.

The DNA duplex is stabilized largely by pi-pi interactions between aromatic rings of the bases. This is termed (4)

a) base stacking.

b) base stacking.

(1) A-T, G-C; (2) A-T, G-C; (3) 5, 10, 15, 20

Use the dropdown options to complete the sentences describing the energy of base pairing.

If a(n) [Select an option] base pair with two hydrogen bonds has an energy of $\sim 10\ \frac{kJ}{mole}$, then the the energy of a [Select an option] base pair is ~ [Select an option] This difference in base-pairing energy is often cited as the reason that A/T rich regions of DNA melt more readily than G/C rich regions. But a large contributor to the 'strength' of G/C rich regions is stabilizing energy from base stacking. Consecutive G/C basepairs have a stacking energy of $\sim 45\ \frac{kJ}{mole}$. Consecutive A/T basepairs have a stacking energy of $\sim 20\ \frac{kJ}{mole}$.

(1) A-T, G-C; (2) A-T, G-C; (3) 5, 10, 15, 20

If a(n) A-T ✔ base pair with two hydrogen bonds has an energy of $\sim 10\ \frac{kJ}{mole}$, then the the energy of a G-C ✔ base pair is ~ 15 kJ/mole. ✔ This difference in base-pairing energy is often cited as the reason that A/T rich regions of DNA melt more readily than G/C rich regions. But a large contributor to the 'strength' of G/C rich regions is stabilizing energy from base stacking. Consecutive G/C basepairs have a stacking energy of $\sim 45\ \frac{kJ}{mole}$. Consecutive A/T basepairs have a stacking energy of $\sim 20\ \frac{kJ}{mole}$.

Considering the structure of an **individual deoxynucleoside triphosphate**, identify the following deoxyribose carbons. For each question, enter your answer as a single number; do not include any additional symbols in your responses. If you are unsure of how these carbons are numbered, go back to the DNA structure video. Remember that in molecular structures, oxygen atoms are depicted as red spheres and nitrogen atoms are depicted as blue spheres.

- Which carbon is attached to the nitrogenous base?

  ′

- Which carbon has a hydroxyl group that will interact with a new deoxynucleotide?

  ′

- Which carbon is directly bound to the phosphate group?

  ′

- Which carbon would have an additional hydroxyl group if the deoxyribose is replaced with a ribose?

  ′

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- Which carbon is attached to the nitrogenous base?

  1 ′ ✔

- Which carbon has a hydroxyl group that will interact with a new deoxynucleotide?

  3 ′ ✔

- Which carbon is directly bound to the phosphate group?

  5 ′ ✔

- Which carbon would have an additional hydroxyl group if the deoxyribose is replaced with a ribose?

  2 ′ ✔

Introduction to DNA Replication

The DNA polymerase found in *E. coli* cells, similar to that of *Drosophila melanogaster*, synthesizes DNA extremely quickly. The *E. coli* polymerase synthesizes around 1000 nucleotides per second. How many seconds will it take one DNA polymerase to synthesize one strand of the *E. coli* genome (4.6×10^6 base pairs)? Do not include any commas in your answer.

Introduction to DNA Replication

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4600 seconds ✔

Introduction to DNA Replication

E. coli cells actually replicate the genome in about 40 minutes. What does this tell you about the number of DNA polymerases replicating the genome of an *E. coli* cell at a given time point? Any thoughtful answer will receive credit; click the Show Answer button to check your thoughts.

Introduction to DNA Replication

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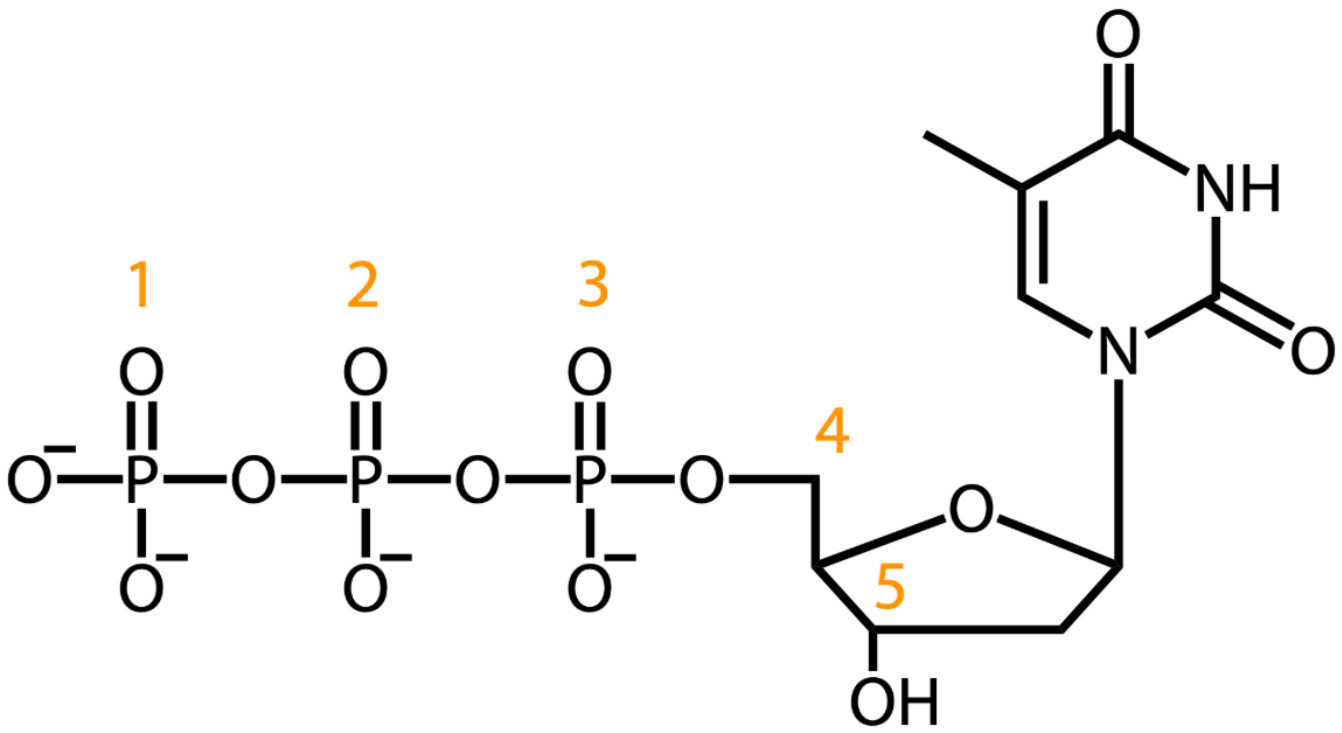
2 ✔

Explanation

There must be more than one DNA polymerase synthesizing the genome at one time during DNA replication of the genome. According to our calculation in Part A, one DNA polymerase can replicate the *E. coli* genome in about 80 minutes. Therefore, two DNA polymerases should be able replicate the *E. coli* genome in about half the time, 40 minutes. We will come back to this concept and the idea that *E.coli* can divide as rapidly as every 20 minutes when we discuss origins of replication later in the course.

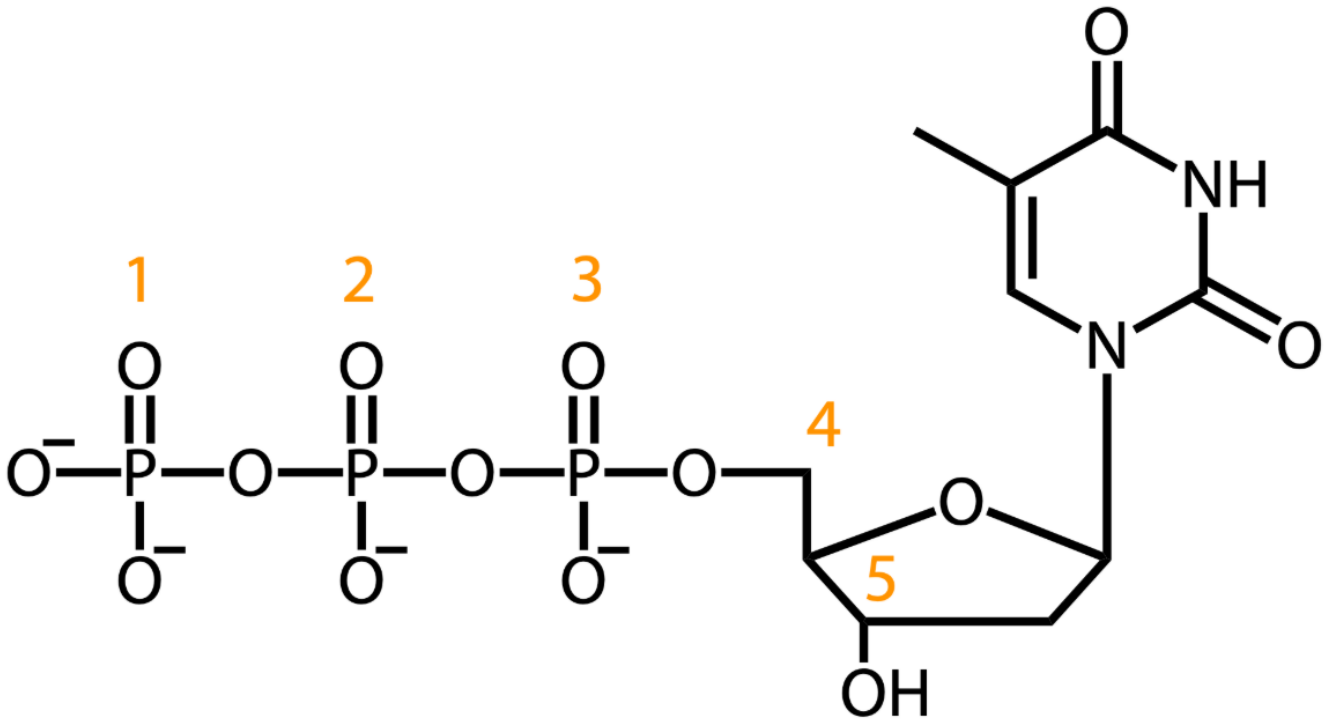
The Chemistry of DNA Replication

Using the dropdown options, identify each phosphate group, the specified carbons, and which nucleotide is depicted in the following image.



- Which number in the structure corresponds to the alpha (α) phosphate? Select an option
- Which number in the structure corresponds to the beta (β) phosphate? Select an option
- Which number in the structure corresponds to the gamma (γ) phosphate? Select an option
- Which number in the structure corresponds to the $3'$ carbon? Select an option
- Which number in the structure corresponds to the $5'$ carbon? Select an option
- Which nucleotide is depicted? Select an option

Using the dropdown options, identify each phosphate group, the specified carbons, and which nucleotide is depicted in the following image.



- Which number in the structure corresponds to the alpha (α) phosphate? 3 ✔
- Which number in the structure corresponds to the beta (β) phosphate? 2 ✔
- Which number in the structure corresponds to the gamma (γ) phosphate? 1 ✔
- Which number in the structure corresponds to the $3'$ carbon? 5 ✔
- Which number in the structure corresponds to the $5'$ carbon? 4 ✔
- Which nucleotide is depicted? dTTP ✔

The DNA Polymerase Active Site

In the mechanism of DNA polymerase catalysis, how does the replicative DNA polymerase ensure that the correct base pair is made? Select the best answer.

1) ◉ accomplishing slow catalysis in the presence of the correct base pair

2) ○ accomplishing fast catalysis in the presence of the correct base pair

3) ○ interacting with and recognizing the correct incoming dNTP

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✔

Thank you for your attention!

- Contact:
  - Email: luu.p.loi@googlemail.com
  - Phone: 0901802182
- For further information